

B.2: Fetal assessment

Background

Assessment of fetal growth and well-being is an important part of ANC. The GDG considered evidence and other relevant information on the following interventions to assess fetal growth and well-being in healthy pregnant women not at risk of adverse perinatal outcomes:

■ **Daily fetal movement counting:** Maternal perception of reduced fetal movements is associated with poor perinatal outcomes, including fetal death (106). Daily fetal movement counting, such as the Cardiff “count-to-ten” method using kick charts, is a way of screening for fetal well-being, by which a woman counts daily fetal movements to assess the condition of her baby. The aim of this is to try to reduce perinatal mortality by alerting health workers when the baby might be compromised (107). Daily fetal movement counting may be used routinely in all pregnant women or only in women who are considered to be at increased risk of adverse perinatal outcomes. Early detection of fetal compromise could lead to timely clinical interventions to reduce poor perinatal outcomes but might lead to maternal anxiety or unnecessary clinical interventions. It is also possible that the period between decreased fetal movements and fetal death might be too short to allow effective action to be taken (108).

■ **Symphysis-fundal height (SFH) measurement:** SFH measurement is a commonly-practiced method of fetal growth assessment that uses a tape measure to measure the SFH, in order to detect intrauterine growth restriction (IUGR). It also has the potential to detect multiple pregnancy, macrosomia, polyhydramnios and oligohydramnios. For fetuses growing normally, from 24 weeks of gestation, the SFH measurement in centimetres should correspond to the number of weeks of gestation, with an allowance of a 2-cm difference either way (109). Other methods of fetal growth assessment include abdominal palpation of fundal height in relation to anatomical landmarks such as the umbilicus and xiphisternum, abdominal girth measurement, and serial ultrasound measurement of the fetal parameters (109). Accurate low-cost methods for detecting abnormal growth are desirable because ultrasound, the most

accurate screening tool, is resource-intensive and not widely available in LMICs.

■ **Routine antenatal cardiotocography (CTG):**

CTG is a continuous recording of the fetal heart rate and uterine contractions obtained via an ultrasound transducer placed on the mother’s abdomen. CTG is widely used in pregnancy as a method of assessing fetal well-being, predominantly in pregnancies with increased risk of complications and during labour.

■ **Fetal ultrasound examination:** Diagnostic ultrasound examination is employed in a variety of specific circumstances during pregnancy, such as where there are concerns about fetal growth and after clinical complications. However, because adverse outcomes may also occur in pregnancies without clear risk factors, assumptions have been made that antenatal ultrasound examination in all pregnancies will prove beneficial by enabling earlier detection of problems that may not be apparent (110) – such as multiple pregnancies, IUGR, congenital anomalies, malpresentation and placenta praevia – and by allowing accurate gestational age estimation, leading to timely and appropriate management of pregnancy complications.

■ **Fetal Doppler ultrasound examination:** Doppler ultrasound technology evaluates umbilical artery (and other fetal arteries) waveforms to assess fetal well-being in the third trimester of pregnancy. It is widely used in high-risk pregnancies to identify fetal compromise and thus reduce perinatal mortality (111, 112). Therefore, it might also be useful when performed as an antenatal intervention to detect fetal compromise and predict complications, particularly IUGR and pre-eclampsia, in apparently healthy pregnancies. Doppler ultrasound is useful for distinguishing between fetuses that are growth-restricted (IUGR) and those that are constitutionally small (SGA) (113). It can be performed as part of a fetal ultrasound examination or separately. The examination quantifies blood flow through the umbilical artery as either a pulsatility index or a resistive index (114). A high resistance to blood flow often indicates an increased risk of IUGR and pre-eclampsia and indicates the need for further investigation.

Women's values

A scoping review of what women want from ANC and what outcomes they value informed the ANC guideline (13). Evidence showed that women from high-, medium- and low-resource settings valued having a positive pregnancy experience. Within the context of maternal and fetal assessment, women valued the opportunity to receive screening and tests to optimize their health and that of their baby as long as individual procedures were explained to them clearly and administered by knowledgeable, supportive and respectful health-care practitioners (high confidence in the evidence).

B.2.1: Daily fetal movement counting

RECOMMENDATION B.2.1: Daily fetal movement counting, such as with “count-to-ten” kick charts, is only recommended in the context of rigorous research. (*Context-specific recommendation – research*)

Remarks

- Fetal movement counting is when a pregnant woman counts and records her baby's movements in order to monitor the baby's health. Various methods have been described, with further monitoring variously indicated depending on the method used, for example, if fewer than six distinct movements are felt within 2 hours (115) or fewer than 10 distinct movements are felt within 12 hours (the Cardiff “count to ten” method) (106).
- While daily fetal movement counting is not recommended, healthy pregnant women should be made aware of the importance of fetal movements in the third trimester and of reporting reduced fetal movements.
- Clinical enquiry by ANC providers at each ANC visit about maternal perception of fetal movements is recommended as part of good clinical practice. Women who perceive poor or reduced fetal movements require further monitoring (e.g. with daily fetal movement counting) and investigation, if indicated.
- The GDG agreed that more research is needed on the effects of daily fetal movement counting in the third trimester of pregnancy, particularly in LMIC settings with a high prevalence of unexplained stillbirths.

Summary of evidence and considerations

Effects of daily maternal fetal movement counting compared with standard ANC (EB Table B.2.1)

The evidence on the effects of daily fetal movement counting was derived from a Cochrane review (107). Two RCTs from HICs contributed data for this comparison. One was a large, multicentre, cluster RCT (68 654 women) conducted in Belgium, Ireland, Sweden, the United Kingdom and the USA, which compared a “count-to-ten” fetal movement counting kick chart with standard ANC in women with uncomplicated pregnancies recruited between 28 and 32 weeks of gestation. Women in the standard ANC group were asked about fetal movements at each ANC visit. The other trial was a multicentre RCT conducted in Norway involving 1123 women that

compared a modified “count-to-ten” fetal movement counting protocol with standard care.

Maternal outcomes

Low-certainty evidence suggests that daily fetal movement counting may make little or no difference to caesarean section (1 trial, 1076 women; RR: 0.93, 95% CI: 0.60–1.44) or assisted vaginal delivery rates (1 trial, 1076 women; RR: 1.04, 95% CI: 0.65–1.66).

With regard to maternal satisfaction, low-certainty evidence suggests that daily fetal movement counting may reduce mean anxiety scores (1 trial, 1013 women; standardized MD: –0.22, 95% CI: –0.35 to –0.10).

Fetal and neonatal outcomes

Low-certainty evidence suggests that there may be little or no difference to preterm birth (1 trial, 1076

neonates; RR: 0.81, 95% CI: 0.46–1.46) and low birth weight (1 trial, 1076 neonates; RR: 0.98, 95% CI: 0.66–1.44) with daily fetal movement counting.

There were no perinatal deaths in the Norwegian trial (1076 women). Low-certainty evidence from the large cluster RCT, which reported the weighted mean difference in stillbirth rates between intervention and control clusters, suggests that fetal movement counting may make little or no difference to stillbirth rates (weighted MD: 0.23, 95% CI: –0.61 to 1.07).

Additional considerations

- These trials were conducted in HICs with low stillbirth rates, therefore the findings on effects may not apply equally to settings with high stillbirth rates.
- In the cluster RCT, despite fetal movement counting, most fetuses detected as being compromised by reduced fetal movements had died by the time the mothers received medical attention.
- There was a trend towards increased CTG and antenatal hospital admissions in the intervention clusters of the cluster RCT. Antenatal hospital admissions were also more frequent in the intervention arm of the Norwegian RCT (107).
- Findings from an additional RCT that was unpublished at the time of the Cochrane review support the Cochrane evidence that daily fetal movement counting may reduce maternal anxiety (115).

Values

Please see “Women’s values” in section 3.B.2: Fetal assessment: Background (p. 54).

Resources

Fetal movement counting is a low-cost intervention on its own, but it could be resource-intensive if it leads to unnecessary additional interventions or hospital admissions.

Equity

LMICs bear the global burden of perinatal morbidity and mortality, and women who are poor, least educated and residing in rural areas of LMICs have lower ANC coverage and worse pregnancy outcomes than more advantaged women (29). Therefore, simple, effective, low-cost antenatal interventions to assess fetal well-being could help to address health inequalities by improving detection of complications in low-resource settings.

Acceptability

Qualitative evidence shows that women generally appreciate the knowledge and information they can acquire from health-care providers during ANC visits, provided this is explained properly and delivered in a consistent, caring and culturally sensitive manner (high confidence in the evidence) (22). It also shows that health professionals want to give appropriate information and advice to women but sometimes they don’t feel suitably trained to do so (high confidence in the evidence) (45).

Feasibility

From the perspective of women who live far from ANC clinics and who may not have the resources or time to attend ANC regularly, and the perspective of ANC providers with limited resources, this intervention may offer a practical and cost-effective approach to monitoring fetal well-being if it’s shown to be effective (high confidence in the evidence) (22, 45).

B.2.2: Symphysis-fundal height (SFH) measurement

RECOMMENDATION B.2.2: Replacing abdominal palpation with symphysis-fundal height (SFH) measurement for the assessment of fetal growth is not recommended to improve perinatal outcomes. A change from what is usually practiced (abdominal palpation or SFH measurement) in a particular setting is not recommended. (Context-specific recommendation)

Remarks

- SFH measurement is routinely practiced in many ANC settings. Due to a lack of clear evidence of accuracy or superiority of either SFH measurement or clinical palpation to assess fetal growth, the GDG does not recommend a change of practice.
- The GDG agreed that there is a lack of evidence on SFH, rather than a lack of effectiveness, particularly in LMIC settings.
- Apart from false reassurance, which might occur with both SFH measurement and clinical palpation, there is no evidence of harm with SFH measurement.
- Research is needed to determine the role of SFH measurement in detecting abnormal fetal growth and other risk factors for perinatal morbidity (e.g. multiple pregnancy, polyhydramnios) in settings where antenatal ultrasound is not available.

Summary of evidence and considerations

Effects of SFH measurement versus abdominal palpation (EB Table B.2.2)

The evidence on the effects of SFH measurement was derived from a Cochrane review that included only one trial conducted in Denmark involving 1639 pregnant women enrolled at about 14 weeks of gestation (109). SFH measurement or abdominal palpation were performed from 28 weeks of gestation. Most women had at least three assessments, with measurements plotted on a chart.

Maternal outcomes

Low-certainty evidence suggests that there may be little or no difference in the effect of SFH measurement versus clinical palpation on caesarean section (1639 women; RR: 0.72, 95% CI: 0.31–1.67) and induction of labour (1639 women; RR: 0.84, 95% CI: 0.45–1.58).

Fetal and neonatal outcomes

Moderate-certainty evidence shows that SFH measurement versus clinical palpation probably makes little or no difference to the antenatal detection of SGA neonates (1639 women; RR: 1.32, 95% CI: 0.92–1.90) and low-certainty evidence suggests that it may make little or no difference to perinatal mortality (1639 women; RR: 1.25, 95% CI: 0.38–4.07). No other ANC guideline outcomes were reported in the review.

Additional considerations

- The GDG also considered evidence from a test accuracy review regarding the accuracy of SFH

in predicting SGA at birth (birthweight < 10th centile), where SGA was a proxy outcome for IUGR (116). The DTA review included seven studies conducted in HICs, which used different measurement thresholds to detect SGA. SFH measurement had a sensitivity ranging from 0.27 to 0.76, suggesting that it fails to identify up to 73% of pregnancies affected by SGA at birth. However, there was generally a high degree of specificity (0.79–0.92), suggesting that a normal SFH measurement may be a reasonable indicator of a healthy baby. In practice, this could mean that few healthy pregnancies are referred for ultrasound examination; however, most true SGA cases may be missed. Comparable test accuracy evidence on abdominal palpation is not available.

Values

Please see “Women’s values” in section 3.B.2: Fetal assessment: Background (p. 54).

Resources

Both abdominal palpation and SFH measurement are low-cost interventions with the main cost being staff training. SFH requires tape measures to be available.

Equity

LMICs bear the global burden of perinatal morbidity and mortality, and women who are poor, least educated and residing in rural areas of LMICs have lower ANC coverage and worse pregnancy outcomes than more advantaged women (29). Therefore, simple, effective, low-cost, routine antenatal

interventions to assess fetal well-being could help to address health inequalities by improving detection of complications in low-resource settings.

Acceptability

SFH and clinical palpation are non-invasive approaches for fetal assessment, which are widely used and not known to be associated with acceptability issues.

However, in some settings women experience a sense of shame during physical examinations, and this needs to be addressed with sensitivity by health-care providers (low confidence in the evidence) (22).

Feasibility

Both methods are considered equally feasible, provided tape measures are available.

B.2.3: Antenatal cardiotocography (CTG)

RECOMMENDATION B.2.3: Routine antenatal cardiotocography is not recommended for pregnant women to improve maternal and perinatal outcomes. (Not recommended)

Remarks

- CTG is the continuous recording of the fetal heart rate and uterine contractions obtained via an ultrasound transducer placed on the mother's abdomen.
- There is currently no evidence on effects or other considerations that supports the use of antenatal (prelabour) CTG as part of routine ANC.
- A lack of evidence of benefits associated with CTG in high-risk pregnancies suggests that the evaluation of antenatal CTG in healthy pregnant women is not a research priority.

Summary of evidence and considerations

Effects of routine antenatal CTG versus no routine antenatal CTG (EB Table B.2.3)

A Cochrane review of routine antenatal CTG for fetal assessment identified no eligible studies of routine CTG and all six included studies involved women with high-risk pregnancies (117).

Additional considerations

- Low-certainty evidence on antenatal CTG in high-risk pregnancies suggests that this intervention may have little or no effect on perinatal mortality and caesarean section (117).

Values

Please see "Women's values" in section 3.B.2: Fetal assessment: Background (p. 54).

Resources

CTG machines are costly (starting from about US\$ 450)⁴, require maintenance and supplies of ultrasound gel, and require staff training in their use and interpretation.

Equity

Simple, effective, low-cost, antenatal interventions to assess fetal well-being could help to address health inequalities by improving detection of complications in low-resource settings, which bear the burden of perinatal mortality.

Acceptability

Qualitative evidence from a variety of settings indicates that women generally appreciate the use of technology to monitor pregnancy (high confidence in the evidence), and a lack of modern equipment at ANC facilities in LMICs may discourage women from attending (moderate confidence in the evidence) (22). However, in some LMICs, women hold the belief that pregnancy is a healthy condition and may be resistant to CTG use unless they have experienced a previous pregnancy complication (high confidence in the evidence). Acceptability may be further compromised if the reasons for using CTG are not properly explained (high confidence in the evidence).

Feasibility

Health-care providers in LMIC settings feel that a lack of modern equipment and training limits the implementation of this type of intervention (high confidence in the evidence) (45).

4 Crude estimate based on Internet search.

B.2.4: Ultrasound scan

RECOMMENDATION B.2.4: One ultrasound scan before 24 weeks of gestation (early ultrasound) is recommended for pregnant women to estimate gestational age, improve detection of fetal anomalies and multiple pregnancies, reduce induction of labour for post-term pregnancy, and improve a woman's pregnancy experience. (Recommended)

Remarks

- The benefits of an early ultrasound scan are not improved upon and cannot be replicated with a late ultrasound scan where there has not been an early ultrasound scan. Therefore, an ultrasound scan after 24 weeks of gestation (late ultrasound) is not recommended for pregnant women who have had an early ultrasound scan. However, stakeholders should consider offering a late ultrasound scan to pregnant women who have not had an early ultrasound scan, for the purposes of identifying the number of fetuses, presentation and placental location.
- The GDG noted that the effects of introducing antenatal ultrasound on population health outcomes and health systems in rural, low-resource settings are unproven. However, the introduction of ultrasound to detect pregnancy complications and confirm fetal viability to the woman and her family in these settings could plausibly increase ANC service utilization and reduce morbidity and mortality, when accompanied by appropriate gestational age estimation, diagnosis, referral and management.
- The ongoing multicountry trial that is under way should contribute further evidence on health effects, health care utilization and implementation-related information on ultrasound in rural, low-resource settings (118).
- The GDG acknowledged that the use of early pregnancy ultrasound has not been shown to reduce perinatal mortality. The GDG put emphasis on other benefits of ultrasound (mentioned in points above) and the increased accuracy of gestational age assessment, which would assist management in case of suspected preterm birth and reduce labour induction for post-term pregnancies.
- The GDG acknowledges that implementing and scaling up this recommendation in low-resource settings will be associated with a variety of challenges that may include political (budgeting for fees and tariffs), logistical (equipment maintenance, supplies, technical support), infrastructural (ensuring a reliable power supply and secure storage) and resources.
- The GDG noted that antenatal ultrasound is an intervention that can potentially be task shifted from trained sonographers and doctors to trained nurses, midwives and clinical officers, provided that ongoing training, staff retention, quality improvement activities and supervision are ensured.
- Stakeholders might be able to offset/reduce the cost of antenatal ultrasound if the ultrasound equipment is also used for other indications (e.g. obstetric emergencies) or by other medical departments.
- The implementation and impact of this recommendation on health outcomes, facility utilization and equity should be monitored at the health service, regional and country levels, based on clearly defined criteria and indicators associated with locally agreed targets.^a
- For further guidance, please refer to the WHO *Manual of diagnostic ultrasound* (119), available at: http://www.who.int/medical_devices/publications/manual_ultrasound_pack1-2/en/

^a Two members of the GDG (Lisa Noguchi and Charlotte Warren) indicated that they would prefer to recommend this intervention in specific contexts with capacity to conduct close monitoring and evaluation to ensure a basic standard of implementation (including adequate capacity to diagnose and manage complications) and monitor for potential adverse effects on delivery of other critical maternal and newborn health interventions.

Summary of evidence and considerations

a) Effects of an ultrasound scan before 24 weeks of gestation (early ultrasound scan) versus selective ultrasound scan (EB Table B.2.4a)

The evidence on early ultrasound was derived from a Cochrane review that included 11 RCTs conducted in Australia, Norway, South Africa, Sweden, the United Kingdom and the USA, involving 37 505

women (120). The intervention in all trials involved an ultrasound scan before 24 weeks of gestation, with women in the control arm undergoing selective scans if indicated (or, in one study, concealed scans, the results of which were not shared with clinicians unless requested). The scans usually included assessment of gestational age (biparietal diameter with or without head circumference and femur length), fetal anatomy, number of fetuses and

location of the placenta. Scans were performed in most trials between 10 and 20 weeks of gestation, with three trials evaluating scans before 14 weeks, and three trials evaluating an intervention comprising both early (at 18–20 weeks) and late scans (at 31–33 weeks).

Maternal outcomes

Moderate-certainty evidence suggests that an early ultrasound scan probably has little or no effect on caesarean section rates (5 trials, 22 193 women; RR: 1.05; 95% CI: 0.98–1.12). However, low-certainty evidence suggests that early ultrasound may lead to a reduction in induction of labour for post-term pregnancy (8 trials, 25 516 women; RR: 0.59, 95% CI: 0.42–0.83).

Regarding maternal satisfaction, low-certainty evidence suggests that fewer women may report feeling worried about their pregnancy after an early ultrasound scan (1 trial, 635 women; RR: 0.80, 95% CI: 0.65–0.99).

Fetal and neonatal outcomes

Low-certainty evidence suggests that early ultrasound scans may increase the detection of congenital anomalies (2 trials, 17 158 women; RR: 3.46, 95% CI: 1.67–7.14). However, detection rates were low for both groups (16% vs 4%, respectively) with 346/387 neonates with abnormalities (89%) being undetected by 24 weeks of gestation.

Low-certainty evidence suggests that early ultrasound may make little or no difference to perinatal mortality (10 trials, 35 737 births; RR: 0.89, 95% CI: 0.70–1.12) and low birth weight (4 trials, 15 868 neonates; RR: 1.04, 95% CI: 0.82–1.33). Moderate-certainty evidence also shows that it probably has little or no effect on SGA (3 trials, 17 105 neonates; RR: 1.05, 95% CI: 0.81–1.35).

b) Effects of an ultrasound scan after 24 weeks of gestation (late ultrasound scan) versus no late ultrasound scan (EB Table B.2.4b)

This evidence on late ultrasound was derived from a Cochrane review that included 13 RCTs conducted in HICs (121). Most women in these trials underwent early ultrasound scan and were randomized to receive an additional third trimester scan or to selective or concealed ultrasound scan. The purpose of the late scan in these trials, which was usually performed between 30 and 36 weeks of gestation, variably included assessment of fetal anatomy, estimated

weight, amniotic fluid volume and/or placental maturity.

Maternal outcomes

Moderate-certainty evidence suggests that a late ultrasound scan probably has little or no effect on caesarean section (6 trials, 22 663 women; RR: 1.03, 95% CI: 0.92–1.15), instrumental delivery (5 trials, 12 310 women; RR: 1.05, 95% CI: 0.95–1.16) and induction of labour (6 trials, 22 663 women; RR: 0.93, 95% CI: 0.81–1.07). Maternal satisfaction was not assessed in this review.

Fetal and neonatal outcomes

Moderate-certainty evidence suggests that a late ultrasound scan probably has little or no effect on perinatal mortality (8 trials, 30 675 births; RR: 1.01, 95% CI: 0.67–1.54) and preterm birth (2 trials, 17 151 neonates; RR: 0.96, 95% CI: 0.85–1.08). Low-certainty evidence suggests that it may have little or no effect on SGA (4 trials, 20 293 neonates; RR: 0.98, 95% CI: 0.74–1.28) and low birth weight (3 trials, 4510 neonates; RR: 0.92, 95% CI: 0.71–1.18).

Additional considerations

- The evidence on ultrasound is derived mainly from HICs, where early ultrasound is a standard component of ANC to establish an accurate gestational age and identify pregnancy complications. The impact of ultrasound screening in low-resource settings is currently unknown but the low rates of maternal and perinatal mortality experienced in HICs indirectly suggests that ultrasound is an important component of quality ANC services.
- Evidence from the Cochrane review on early ultrasound suggests that multiple pregnancies may be less likely to be missed/undetected by 24–26 weeks of gestation with early ultrasound (120). Of 295 multiple pregnancies occurring in seven trials (approximately 24 000 trial participants), 1% (2/153) were undetected by 24–26 weeks of gestation with early ultrasound screening compared with 39% (56/142) in the control group (RR: 0.07, 95% CI: 0.03–0.17; graded by review authors as low-quality evidence).
- The Cochrane review also evaluated several safety outcomes in offspring and found no evidence of differences in school performance, vision and hearing, disabilities or dyslexia.
- An ongoing multicountry cluster RCT of antenatal ultrasound in the Democratic Republic of the Congo, Guatemala, Kenya, Pakistan and Zambia

should contribute data on health outcomes and health care utilization, as well as implementation-related information on ultrasound in rural, low-resource settings (118). The trial intervention involves a two-week obstetric ultrasound training course for health workers (e.g. midwives, nurses, clinical officers) to perform ultrasound scans at 18–22 weeks and 32–36 weeks of gestation in each participant enrolled.

- Accurate gestational age dating is critical for the appropriate delivery of time-sensitive interventions in pregnancy, as well as management of pregnancy complications, particularly pre-eclampsia and preterm birth, which are major causes of maternal and perinatal morbidity and mortality in LMICs, and early ultrasound is useful for this purpose.

Values

Please see “Women’s values” in section 3.B.2: Fetal assessment: Background (p. 54).

Resources

The cost of ultrasound equipment, especially portable compact units, has decreased (122), and they are currently available at less than US\$ 10 000 (28). Thus, given the cost of equipment, maintenance, supplies (ultrasound gel), replacement batteries, initial and ongoing staff training and supervision, and staffing costs (allowing 15–45 minutes per scan), routine ultrasound scans may have considerable resource implications for LMIC settings.

Equity

Effective interventions to increase uptake and quality of ANC services, and improve the experience of care, are needed in LMICs to prevent maternal and perinatal mortality and improve equity. However, if women are expected to pay for ultrasound scans, or if scans are not available to women living in rural areas due to feasibility issues, this intervention could perpetuate inequalities. In addition, ultrasound sexing of the fetus in some low-income countries has a negative impact on gender equity and needs to be monitored.

Acceptability

Qualitative evidence shows that women generally appreciate the knowledge and information they can acquire from health-care providers and that they are willing to be screened and tested for a variety of conditions, provided the information and procedures are explained properly and delivered in a caring and culturally sensitive manner (high confidence in the evidence) (22). Evidence also shows that, in some LMICs, the lack of modern technology (like ultrasound equipment) at ANC facilities discourages some women from attending (high confidence in the evidence) (22). This suggests that the offer of ultrasound might attract women to use ANC facilities, which may also lead to earlier ANC attendance. Specific studies not included in the main qualitative review indicate that women value the opportunity to see their baby via ultrasound and find the test reassuring (123). However, there is some evidence that women do not understand that ultrasound is a diagnostic tool, and that adverse findings during scans might increase anxiety and distress (124).

Qualitative evidence from health-care providers shows that they generally want to provide screening and testing procedures, but sometimes don’t feel suitably trained to do so (high confidence in the evidence) (45). This suggests that they might welcome ultrasound scans to assist with accurate gestational age estimation and to identify potential risk factors, such as multiple pregnancies, if appropriately trained and supported.

Feasibility

Feasibility challenges of antenatal ultrasound scans in LMICs includes equipment procurement and staff training, ensuring a power supply (via a power point or rechargeable batteries) and secure storage, regular equipment maintenance, maintaining adequate and continual supplies of ultrasound gel, and ongoing technical support and supervision.

B.2.5: Doppler ultrasound of fetal blood vessels

RECOMMENDATION B.2.5: Routine Doppler ultrasound examination is not recommended for pregnant women to improve maternal and perinatal outcomes. (*Not recommended*)

Remarks

- The GDG noted that the evidence base for the use of Doppler ultrasound of fetal blood vessels in high-risk pregnancy is already established.
- The GDG agreed that the value of a single Doppler ultrasound examination of fetal blood vessels for all pregnant women in the third trimester needs rigorous evaluation, particularly in LMIC settings. Future trials should be designed to evaluate the effect of a single Doppler ultrasound on preventable perinatal deaths.

Summary of evidence and considerations

Effects of Doppler ultrasound examination of fetal blood vessels compared with no Doppler ultrasound examination (EB Table B.2.5)

The evidence on Doppler ultrasound examination was derived from a Cochrane review that included five trials involving 14 624 women in HICs (Australia, France and the United Kingdom) (114). One study evaluated a single Doppler examination at 28–34 weeks of gestation, three studies evaluated multiple Doppler examinations from as early as 18 weeks, and one study evaluated women undergoing single or multiple examinations from 26 to 36 weeks of gestation. Data were evaluated together and separately for single and multiple examinations. Women in the control arms received standard ANC with no (or concealed) Doppler examination.

Maternal outcomes

The available moderate-certainty evidence suggests that antenatal Doppler ultrasound probably makes little or no difference to caesarean section rates (2 trials, 6373 women; RR: 0.98, 95% CI: 0.85–1.13) and assisted vaginal birth (2 trials, 6884 women; RR: 1.04, 95% CI: 0.96–1.12). No other maternal outcomes that were prioritized for the ANC guideline were reported in the trials.

Fetal and neonatal outcomes

Low-certainty evidence suggests that Doppler ultrasound may have little or no effect on perinatal mortality (4 trials, 11 183 women; RR: 0.80, 95% CI: 0.35–1.83). Moderate-certainty evidence indicates that the intervention probably has little or no effect on preterm birth (4 trials, 12 162 women; RR: 1.02, 95% CI: 0.87–1.18).

Additional considerations

- Subgroup analyses according to the number of Doppler ultrasound examinations (single or multiple) are largely consistent with the overall findings. However, low-certainty evidence from the single examination subgroup suggests that a single Doppler ultrasound examination might reduce perinatal mortality (1 trial, 3890 women; RR: 0.36, 95% CI: 0.13–0.99).

Values

Please see “Women’s values” in section 3.B.2: Fetal assessment: Background (p. 54).

Resources

The cost of ultrasound equipment, especially portable compact units, has decreased (122), and they are currently available at less than US\$ 10 000 (28). Thus, given the cost of equipment, maintenance, supplies (ultrasound gel), replacement batteries, initial and ongoing staff training and supervision, and staffing costs, routine Doppler ultrasound scans may have considerable resource implications for LMIC settings.

Equity

RCT evidence on maternal and perinatal effects of Doppler ultrasound examination is currently derived from HICs and high-quality research is needed on this intervention in LMICs to determine whether, by improving detection of pregnancy complications, it can reduce perinatal mortality and improve health equity.

Acceptability

Qualitative evidence shows that women generally appreciate the knowledge and information they can acquire from health-care providers and that they